KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5111 | 5 Credits: 4 | Program Core | Source-grounded

Tool Design

 To impart basic knowledge of kinetic link mechanism and degrees of freedom

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5111 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5111.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Tool & Die Engineering
Course code	5111
Course title	Tool Design
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart basic knowledge of kinetic link mechanism and degrees of freedom
- To familiarize students with design techniques of press tools and moulds
- To provide the basic understanding of jigs and fixtures used in tool making industry

Course outcomes

CO	Official outcome	Study level
CO1	elements, basics of Kinetic link, design of 15 Understanding threaded fasteners and basics of die Design Apply the concept of various die design and its	See official syllabus
CO2	18 Applying components Explain the concept of various mould design and	See official syllabus
CO3	14 Understanding its components Recognize various Jigs and Fixtures, locators and	See official syllabus
CO4	11 Understanding clamps	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Kinetic link, design of threaded fasteners and basics of die Design	4	As specified
Module 2	Apply the concept of die design and its components	4	As specified
Module 3	Explain the concept of mould design and its components	4	As specified
Module 4	Recognize suitable Jigs and Fixtures, locators and clamps	4	As specified

MODULE 1

Kinetic link, design of threaded fasteners and basics of die Design

This module is presented from the official syllabus scope for Tool Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Kinetic link, design of threaded fasteners and basics of die Design
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain the basics of kinetic link and mechanism

Explain the basics of kinetic link and mechanism is an official topic in Module 1 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the basics of kinetic link and mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the basics of kinetic link and mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the basics of kinetic link and mechanism means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the basics of kinetic link and mechanism □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ Familiar with thread type fasteners 3 Understanding Explain the basics of die design and die

Familiar with thread type fasteners 3 Understanding Explain the basics of die design and die is an official topic in Module 1 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiar with thread type fasteners 3 Understanding Explain the basics of die design and die using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiar with thread type fasteners 3 understanding explain the basics of die design and die should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Familiar with thread type fasteners 3 Understanding Explain the basics of die design and die means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Familiar with thread type fasteners 3 Understanding Explain the basics of die design and die □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 4 Understanding components

4 Understanding components is an official topic in Module 1 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding components using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding components should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding components means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	**What to prepare**
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding components □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Describe different die operations 4 Understanding General Considerations in Machine Design Kinetic Link: Basic link motion, Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom Temporary fastener type: Thread: thread type, types of thread, forms of thread, Designation and nomenclature of screw thread Introduction to die design - die components - die operations - Blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming Die classification - Simple Die, Compound Die, Progressive Die, Combination Die

Describe different die operations 4 Understanding General Considerations in Machine Design Kinetic Link: Basic link motion, Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom Temporary fastener type: Thread: thread type, types of thread, forms of thread, Designation and nomenclature of screw thread Introduction to die design - die components - die operations - Blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming Die classification - Simple Die, Compound Die, Progressive Die, Combination Die is an official topic in Module 1 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe different die operations 4 Understanding General Considerations in Machine Design Kinetic Link: Basic link motion, Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom Temporary fastener type: Thread: thread type, types of thread, forms of thread, Designation and nomenclature of screw thread Introduction to die design - die components - die operations - Blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming Die classification - Simple Die, Compound Die, Progressive Die, Combination Die using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe different die operations 4 understanding general considerations in machine design kinetic link: basic link motion, definition and explanation of the terms kinematic link or element, kinematic pair, kinematic chain, mechanism, degrees of freedom temporary fastener type: thread: thread type, types of thread, forms of thread, designation and nomenclature of screw thread introduction to die design - die components - die operations - blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming die classification - simple die, compound die, progressive die, combination die should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe different die operations 4 Understanding General Considerations in Machine Design Kinetic Link: Basic link motion, Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom Temporary fastener type: Thread: thread type, types of thread, forms of thread, Designation and nomenclature of screw thread Introduction to die design - die components - die operations - Blanking, piercing,

Aspect	What to prepare
	shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming Die classification - Simple Die, Compound Die, Progressive Die, Combination Die means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Describe different die operations 4 Understanding General Considerations in Machine Design Kinetic Link: Basic link motion, Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom Temporary fastener type: Thread: thread type, types of thread, forms of thread, Designation and nomenclature of screw thread Introduction to die design - die components - die operations - Blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming Die classification - Simple Die, Compound Die, Progressive Die, Combination Die □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Apply the concept of die design and its components

This module is presented from the official syllabus scope for Tool Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply the concept of die design and its components
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain stepwise design procedure of a die

Explain stepwise design procedure of a die is an official topic in Module 2 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain stepwise design procedure of a die using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain stepwise design procedure of a die should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain stepwise design procedure of a die means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain stepwise design procedure of a die □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

▼ Apply design knowledge of compound tool 4 Applying Apply design knowledge of different types of

Apply design knowledge of compound tool 4 Applying Apply design knowledge of different types of is an official topic in Module 2 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply design knowledge of compound tool 4 Applying Apply design knowledge of different types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply design knowledge of compound tool 4 applying apply design knowledge of different types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Apply design knowledge of compound tool 4 Applying Apply design knowledge of different types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Apply design knowledge of compound tool 4 Applying Apply design knowledge of different types of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ 5 Applying bending tool

5 Applying bending tool is an official topic in Module 2 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying bending tool using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying bending tool should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying bending tool means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Applying bending tool □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Apply the design aspects of drawing tool 5 Applying Blanking force calculation Steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation) Design of Compound tools - With Direct knock out - with indirect knockout. Design of Bending Tools :- V-Bending Tool, U- Bending Tool, L-Bending Tool, curling tool, forming tool, Bending stress - calculation, Bending radius - calculation (simple numerical problems) Design of Drawing tool, Deep draw tool, Inverted draw tool Factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, Blank holding force - calculation.(simple numerical problems)

Apply the design aspects of drawing tool 5 Applying Blanking force calculation Steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation) Design of Compound tools - With Direct knock out - with indirect knockout. Design of Bending Tools :- V-Bending Tool, U-Bending Tool, L-Bending Tool, curling tool, forming tool, Bending stress - calculation, Bending radius - calculation (simple numerical problems) Design of Drawing tool, Deep draw tool, Inverted draw tool Factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, Blank holding force - calculation. (simple numerical problems) is an official topic in Module 2 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply the design aspects of drawing tool 5 Applying Blanking force calculation Steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation) Design of Compound tools - With Direct knock out - with indirect knockout. Design of Bending Tools :- V-Bending Tool, U-Bending Tool, L-Bending Tool, curling tool, forming tool, Bending stress - calculation, Bending radius - calculation (simple numerical problems) Design of Drawing tool, Deep draw tool, Inverted draw tool Factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, Blank holding force - calculation. (simple numerical problems) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply the design aspects of drawing tool 5 applying blanking force calculation steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation) design of compound tools - with direct knock out - with indirect knockout. design of bending tools :- v-bending tool, u- bending tool, l-bending tool, curling tool, forming tool, bending stress - calculation, bending radius - calculation (simple numerical problems) design of drawing tool, deep draw tool, inverted draw tool factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, blank holding force - calculation.(simple numerical problems) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply the design aspects of drawing tool 5 Applying Blanking force calculation Steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation) Design of Compound tools - With Direct knock out - with indirect knockout. Design of Bending Tools :- V-Bending Tool, U- Bending Tool, L-Bending Tool, curling tool, forming tool, Bending stress - calculation, Bending radius - calculation (simple numerical problems) Design of Drawing tool, Deep draw tool, Inverted draw tool Factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, Blank holding force - calculation.(simple numerical problems) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Apply the design aspects of drawing tool 5 Applying Blanking force calculation Steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation) Design of Compound tools - With Direct knock out - with indirect knockout. Design of Bending Tools :- V-Bending Tool, U- Bending Tool, L-Bending Tool, curling tool, forming tool, Bending stress - calculation, Bending radius - calculation (simple numerical problems) Design of Drawing tool, Deep draw tool, Inverted draw tool Factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, Blank holding force - calculation.(simple numerical problems) □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the concept of mould design and its components

This module is presented from the official syllabus scope for Tool Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the concept of mould design and its components
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Describe basics of mould and its terminology 4 Understanding Explain different types of plastic materials and

Describe basics of mould and its terminology 4 Understanding Explain different types of plastic materials and is an official topic in Module 3 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe basics of mould and its terminology 4 Understanding Explain different types of plastic materials and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe basics of mould and its terminology 4 understanding explain different types of plastic materials and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Describe basics of mould and its terminology 4 Understanding Explain different types of plastic materials and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Describe basics of mould and its terminology 4 Understanding Explain different types of plastic materials and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ 2 Understanding its flow properties

2 Understanding its flow properties is an official topic in Module 3 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding its flow properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding its flow properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding its flow properties means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding its flow properties □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Describe stepwise design of injection mould 4 Understanding Demonstrate basic design steps for compression

Describe stepwise design of injection mould 4 Understanding Demonstrate basic design steps for compression is an official topic in Module 3 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe stepwise design of injection mould 4 Understanding Demonstrate basic design steps for compression using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe stepwise design of injection mould 4 understanding demonstrate basic design steps for compression should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Describe stepwise design of injection mould 4 Understanding Demonstrate basic design steps for compression means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Describe stepwise design of injection mould 4 Understanding Demonstrate basic design steps for compression □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ mould, transfer mould, blow mould and die 4 Understanding casting die Definition of mould: Terminology - mould materials - plastic materials - types of plastics- melting - solidification - shrinkage - melt flow index - variable molecular weight - melt temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view Design of Compression moulds: For component showing Horizontal flash, For component showing vertical flash. Design of Transfer mould: Pot transfer mould - plunger transfer mould. Design of die-casting dies: For cold chamber system - for hot chamber system.

mould, transfer mould, blow mould and die 4 Understanding casting die Definition of mould: Terminology - mould materials - plastic materials - types of plastics- melting - solidification - shrinkage - melt flow index - variable molecular weight - melt temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view Design of Compression moulds: For component showing Horizontal flash, For component showing vertical flash. Design of Transfer mould: Pot transfer mould - plunger transfer mould. Design of die-casting dies: For cold chamber system - for hot chamber system. is an official topic in Module 3 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify mould, transfer mould, blow mould and die 4 Understanding casting die Definition of mould: Terminology - mould materials - plastic materials - types of plastics- melting - solidification - shrinkage - melt flow index - variable molecular weight - melt temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view Design of Compression moulds: For component showing Horizontal flash, For component showing vertical flash. Design of Transfer mould: Pot transfer mould - plunger transfer mould. Design of die-casting dies: For cold chamber system - for hot chamber system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mould, transfer mould, blow mould and die 4 understanding casting die definition of mould: terminology - mould materials - plastic materials - types of plastics- melting - solidification - shrinkage - melt flow index - variable molecular weight - melt temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view design of compression moulds: for component showing horizontal flash, for component showing vertical flash. design of transfer mould: pot transfer mould - plunger transfer mould. design of die-casting dies: for cold chamber system - for hot chamber system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What mould, transfer mould, blow mould and die 4 Understanding casting die Definition of mould: Terminology - mould materials - plastic

Aspect	What to prepare
	materials - types of plastics- melting - solidification - shrinkage - melt flow index - variable molecular weight - melt temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view Design of Compression moulds: For component showing Horizontal flash, For component showing vertical flash. Design of Transfer mould: Pot transfer mould - plunger transfer mould. Design of die-casting dies: For cold chamber system - for hot chamber system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ mould, transfer mould, blow mould and die 4 Understanding casting die Definition of mould: Terminology - mould materials - plastic materials - types of plastics- melting - solidification - shrinkage - melt flow index - variable molecular weight - melt temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view Design of Compression moulds: For component showing Horizontal flash, For component showing vertical flash. Design of Transfer mould: Pot transfer mould - plunger transfer mould. Design of die-casting dies: For cold chamber system - for hot chamber system. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□

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▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Recognize suitable Jigs and Fixtures, locators and clamps

This module is presented from the official syllabus scope for Tool Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Recognize suitable Jigs and Fixtures, locators and clamps
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Recognize appropriate Jigs and fixtures

Recognize appropriate Jigs and fixtures is an official topic in Module 4 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recognize appropriate Jigs and fixtures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recognize appropriate jigs and fixtures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Recognize appropriate Jigs and fixtures means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Recognize appropriate Jigs and fixtures □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ Describe design aspect of locators

Describe design aspect of locators is an official topic in Module 4 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe design aspect of locators using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe design aspect of locators should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe design aspect of locators means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe design aspect of locators □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Select suitable clamps

Select suitable clamps is an official topic in Module 4 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Select suitable clamps using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, select suitable clamps should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Select suitable clamps means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Select suitable clamps □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Explain different types of fixtures 3 Understanding Jigs and Fixture: Materials used in Jigs and fixtures Design of locators - principles of degrees of freedom and constraints. Design of clamp: principles and classification of clamps Drilling Jigs: Types - Components Milling fixture: Salient features and classification Other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 Jigs and Fixtures - Joshi T3 Press Tools: Design and Construction - Joshi.P.H. R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Hill R2 Design of Jigs fixtures and press tools - v balachandran R3 The Mould Design Guide - Peter Jones R4 Plastic technology handbook - Donald E Hudgin R5 Jigs and Fixtures - Hiran.E.Grant Online Resources: Sl.No Website Link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://www.misumi-techcentral.com 4 http://www.wisetool.com 5 https://www.thefabricator.com

Explain different types of fixtures 3 Understanding Jigs and Fixture: Materials used in Jigs and fixtures Design of locators - principles of degrees of freedom and constraints. Design of clamp: principles and classification of clamps Drilling Jigs: Types - Components Milling fixture: Salient features and classification Other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 Jigs and Fixtures - Joshi T3 Press Tools: Design and Construction - Joshi.P.H. R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Hill R2 Design of Jigs fixtures and press tools - v balachandran R3 The Mould Design Guide - Peter Jones R4 Plastic technology handbook - Donald E Hudgin R5 Jigs and Fixtures - Hiran.E.Grant Online Resources: Sl.No Website Link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://www.misumi-techcentral.com 4 http://www.wisetool.com 5 https://www.thefabricator.com is an official topic in Module 4 of Tool Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different types of fixtures 3 Understanding Jigs and Fixture: Materials used in Jigs and fixtures Design of locators - principles of degrees of freedom and constraints. Design of clamp: principles and classification of clamps Drilling Jigs: Types - Components Milling fixture: Salient features and classification Other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 Jigs and Fixtures - Joshi T3 Press Tools: Design and Construction - Joshi.P.H. R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Hill R2 Design of Jigs fixtures and press tools - v balachandran R3 The Mould Design Guide - Peter Jones R4 Plastic technology handbook - Donald E Hudgin R5 Jigs and Fixtures - Hiran.E.Grant Online Resources: Sl.No Website Link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://www.misumi-techcentral.com 4 http://www.wisetool.com 5 https://www.thefabricator.com using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different types of fixtures 3 understanding jigs and fixture: materials used in jigs and fixtures design of locators - principles of degrees of freedom and constraints. design of clamp: principles and classification of clamps drilling jigs: types - components milling fixture: salient features and classification other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures text/ reference: t/r book title/author t1 introduction to injection moulding - pye t2 jigs and fixtures - joshi t3 press tools: design and construction - joshi.p.h. r1 fundamentals of plastics mould design - sanjay k nayak - tata mac graw hill r2 design of jigs fixtures and press tools - v balachandran r3 the mould design guide - peter jones r4 plastic technology handbook - donald e hudgin r5 jigs and fixtures - hiran.e.grant online resources: sl.no website link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://www.misumi-techcentral.com 4 http://www.wisetool.com 5 https://www.thefabricator.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different types of fixtures 3 Understanding Jigs and Fixture: Materials used in Jigs and fixtures Design of locators - principles of degrees of freedom and constraints. Design of clamp: principles and classification of clamps Drilling Jigs: Types - Components Milling fixture: Salient features and classification Other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 Jigs and Fixtures - Joshi T3 Press Tools: Design and Construction - Joshi.P.H. R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Hill R2 Design of Jigs fixtures and press tools - v balachandran R3 The Mould Design Guide - Peter Jones R4 Plastic technology handbook - Donald E Hudgin R5 Jigs and Fixtures - Hiran.E.Grant Online Resources: Sl.No Website Link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://www.misumi-techcentral.com 4 http://www.wisetool.com 5 https://www.thefabricator.com means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain different types of fixtures 3 Understanding Jigs and Fixture: Materials used in Jigs and fixtures Design of locators - principles of degrees of freedom and constraints. Design of clamp: principles and classification of clamps Drilling Jigs: Types - Components Milling fixture: Salient features and classification Other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 Jigs and Fixtures - Joshi T3 Press Tools: Design and Construction - Joshi.P.H. R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Hill R2 Design of Jigs fixtures and press tools - v balachandran R3 The Mould Design Guide - Peter Jones R4 Plastic technology handbook - Donald E Hudgin R5 Jigs and Fixtures - Hiran.E.Grant Online Resources: Sl.No Website Link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://www.misumi-techcentral.com 4 http://www.wisetool.com 5 https://www.thefabricator.com □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps →
Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure
→ Observation → Result →
Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
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Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain Explain the basics of kinetic link and mechanism. (3 marks)**

▶ **Define or explain Familiar with thread type fasteners 3 Understanding Explain the basics of die design and die. (3 marks)**

▶ **Define or explain 4 Understanding components. (3 marks)**

▶ **Define or explain Describe different die operations 4 Understanding General Considerations in Machine Design Kinetic Link: Basic link motion, Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom Temporary fastener type: Thread: thread type, types of thread, forms of thread, Designation and nomenclature of screw thread Introduction to die design - die components - die operations - Blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming Die classification - Simple Die, Compound Die, Progressive Die, Combination Die. (3 marks)**

Module 2

▶ **Define or explain Explain stepwise design procedure of a die. (3 marks)**

▶ **Define or explain Apply design knowledge of compound tool 4 Applying Apply design knowledge of different types of. (3 marks)**

▶ **Define or explain 5 Applying bending tool. (3 marks)**

▶ **Define or explain Apply the design aspects of drawing tool 5 Applying Blanking force calculation Steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation) Design of Compound tools - With Direct knock out - with indirect knockout. Design of Bending Tools :- V-Bending Tool, U- Bending Tool, L-Bending Tool, curling tool, forming tool, Bending stress - calculation, Bending radius - calculation (simple numerical problems) Design of Drawing tool, Deep draw tool, Inverted draw tool Factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, Blank holding force - calculation.(simple numerical problems). (3 marks)**

Module 3

▶ **Define or explain Describe basics of mould and its terminology 4 Understanding Explain different types of plastic materials and. (3 marks)**

▶ **Define or explain 2 Understanding its flow properties. (3 marks)**

▶ **Define or explain Describe stepwise design of injection mould 4 Understanding Demonstrate basic design steps for compression. (3 marks)**

▶ **Define or explain mould, transfer mould, blow mould and die 4 Understanding casting die Definition of mould: Terminology - mould materials - plastic materials - types of plastics- melting - solidification - shrinkage - melt flow index - variable molecular weight - melt**

temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view Design of Compression moulds: For component showing Horizontal flash, For component showing vertical flash. Design of Transfer mould: Pot transfer mould - plunger transfer mould. Design of die-casting dies: For cold chamber system - for hot chamber system.. (3 marks)

Module 4

▶ **Define or explain Recognize appropriate Jigs and fixtures. (3 marks)**

▶ **Define or explain Describe design aspect of locators. (3 marks)**

▶ **Define or explain Select suitable clamps. (3 marks)**

▶ **Define or explain Explain different types of fixtures 3 Understanding Jigs and Fixture: Materials used in Jigs and fixtures Design of locators - principles of degrees of freedom and constraints. Design of clamp: principles and classification of clamps Drilling Jigs: Types - Components Milling fixture: Salient features and classification Other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 Jigs and Fixtures - Joshi T3 Press Tools: Design and Construction - Joshi.P.H. R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Hill R2 Design of Jigs fixtures and press tools - v balachandran R3 The Mould Design Guide - Peter Jones R4 Plastic technology handbook - Donald E Hudgin R5 Jigs and Fixtures - Hiran.E.Grant Online Resources: Sl.No Website Link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://www.misumi-techcentral.com 4 http://www.wisetool.com 5 https://www.thefabricator.com. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the basics of kinetic link and mechanism**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain stepwise design procedure of a die**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Describe basics of mould and its terminology 4 Understanding Explain different types of plastic materials and**.

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **Recognize appropriate Jigs and fixtures**.

3-mark explanation: Explain one principle, process or comparison from this module.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://myplasticmold.com/
- https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- https://www.misumi-techcentral.com http://www.wisetool.com https://www.thefabricator.com

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